Eitan Niv

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Robotics, path planning and control engineer with experience deploying real-time, hardware–software systems

EXPERIENCE

Research Scholar **May 2026 – Current**
NYU Center for Robotics, New York City, United States
Developed a quadruped robot traversing and climbing structures.

- Designed and implemented an MPC controller using Pinocchio and Crocoddyl enabling quadruped walking, vertical climbing, and horizontal climbing, after benchmarking it against an IK solver with a custom Whole-Body Controller (WBC).
- Simulated the robot in Mujoco, and continued in deploying the sim-to-real pipeline for the robot.

Robotics Graduate Teaching Assistant - Mechatronics, Dynamics, Math for Robotics and Foundations of Robotics
Sep 2025 – May 2026
NYU Tandon School of Engineering, New York City, United States

- Supported graduate-level robotics coursework covering kinematics, state-space control, estimation (Kalman / BLUE / SVD), and system modeling.

Software Engineer **June 2021 – December 2023**
Zalirian LTD., Tel Aviv, Israel

- Designed and implemented a production-grade, real-time edge control and perception system, integrating motors, industrial cameras, and external hardware SDKs.
- Built high-performance, maintainable software architecture, leveraging multithreading, multiprocessing, design patterns, and C++/Python interoperability (Pybind), with robust logging and telemetry for real-time monitoring and post-deployment support.
- Implemented a Motion Planning algorithm to assure movement efficiency and safety in a 5DoF system with physical limitations and obstacles, thereby improving the movement efficiency and safety.

Software and Hardware Developer **October 2018 – April 2021**
Israeli Air Force, Israel

- Led end-to-end deployment and validation of mission-critical hardware/software systems in live environments.
- Modernized legacy airborne and sensor systems, integrating new hardware with existing infrastructure.
- Applied signal processing to complex data acquisition systems to evaluate reliability, accuracy, and safety.
- Performed system-level fault analysis and debugging to support flight investigations and operational readiness.
- Delivered solutions in environments requiring high reliability, rapid diagnosis, and clear technical documentation.

SKILLS

Software: Python, C++, ROS2, GIT, Pybind, LabVIEW, OpenCV, PyQt, MatLab, Simulink, Nav2, Gazebo, Mujoco, Docker, Pinocchio, Crocoddyl
Technical: Motion Planning, Control Theory, Signal Processing, Sensor and Actuator Integration, Localization and Navigation, EKF, Particle Filtering, Project Management, Client-Development, Requirements analysis, Device Drivers & SDK Integration

PROJECTS

M.Sc Final Project: Designed and implemented a collision mitigation strategy for autonomous vehicles, integrating a multi-stage MPPI controller into the Nav2 library. Developed custom plugins for obstacle surface calculation and created replicable Gazebo testing scenarios and Docker containers.
M.Sc Mechatronics Projects: Programmed a SCARA robot arm that can reproduce intricate sketches using RPi, Arduino and ArUco markers • Simulator for forward and inverse kinematics of a 6DOF robotic arm • Path Planning on UR16e robotic arm in Gazebo using ROS2 • Designed and implemented MPC, MPPI controllers for a simulated drone / robotic arm • Designed a full AMR simulation from dynamics, to motion planning and multiple controllers, using a modular architecture for switching between planners and controllers • Implemented state estimation algorithms using Extended Kalman Filters (EKF) and Particle Filters (PF), in conjunction with either a separate camera or a lidar sensor, given a low quality single encoder to achieve robust and accurate localization on a robotic vehicle.
B.S Final Project: Designing and implementing a self-balancing quadrotor drone control system.

EDUCATION

New York University Tandon School of Engineering **Graduated May 2026**
M.S. in Mechatronics and Robotics – GPA 3.96

Tel Aviv University **Graduated June 2020**
B.S. in Electrical Engineering and Computer Science